



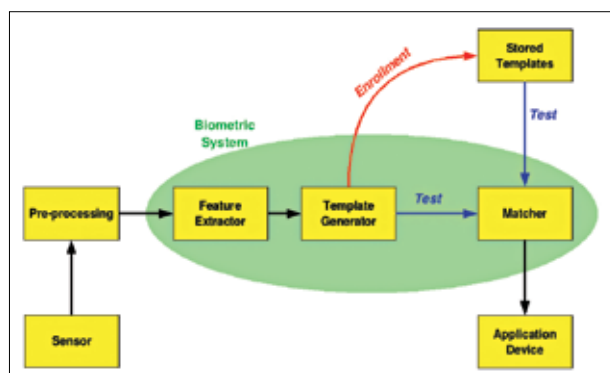
AI finds 'trippy' regions of brain

AI maps 'trip' experiences to regions of the brain, opening new routes to psychiatric treatments. <https://digit.in/apr22-13>

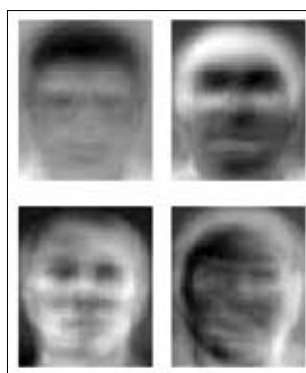


AI beats humans in Bridge

Several world champion bridge players had to accept defeat at the hands of a new AI system. <https://digit.in/apr22-14>



Diagrammatic representation of the working of a basic biometric system



A sample image used in the Eigenface technique

that event, the access systems implemented the use of hand geometry recognition technology successfully and popularised the use of such modern recognition techniques among the masses. Another sporting event, the Super Bowl held in Tampa, Florida, in 2001, saw a relatively less successful implementation of a biometric technique. Here, facial recognition systems had been installed with the motive to identify 'wanted' individuals.

the use of iris-based identification techniques. He examined the patterns of the iris and established that no two iris patterns were the same and thus could be used to attach a unique identity to people. But this idea did not get much momentum until the 1990s when John Daugman developed and patented algorithms to study and recognise iris patterns. Around this time, in the year 1947 to be precise, the First International Biometric Conference was held in Massachusetts.

Meanwhile, other developments raged on. Facial recognition technology, which we all enjoy today in our smartphones, also came to light during this period when semi-automatic facial recognition was developed by Woodrow W. Bledsoe in the 1960s. His technique involved calculations made around the distance between two facial feature reference points. This laid the foundation of a lot of related technologies to be developed in the coming years. Around the same time, the first speech-related identification methods saw some traction building around them when a Swedish professor, Gunnar Fant, published his work around the use of x-rays to identify the speaker of a particular piece of speech.

The FBI of the United States of America was also a major contributor to the development of automated biometric recognition techniques. The FBI, in the late 1960s, pushed for the automation of fingerprint recognition techniques, which led to the development of several subsequent techniques.

North American Aviation in the year 1965 made a major breakthrough, as they developed the first signature recognition system.

All of the developments around this time were restricted to the labs and only special applications. The open commercial markets still awaited the first openly marketable biometric recognition systems after the fingerprint systems, which had been made available in the earlier decade. This came in the form of the first-hand geometry recognition systems, which were launched in the markets in the year 1974.

Following this, the research and development in the field of biometrics picked up pace. With the advent of modern computational techniques coupled with the increasing interest in the field, the world saw the implementation and development of some of the most advanced biometric systems, of the time.

In 1985, the first retinal scanning systems were put to use in the Defence Department of the Naval Postgraduate School in the US. A couple of years later, the patent for identifying the human eye based on the specific features of its iris was awarded to inventors Leonard Flom and Aran Safir. Six years later, facial recognition technology also caught up, and the development of Eigenface technology meant that automated facial recognition was made possible for the first time.

For the first time, biometric techniques saw a widely publicised use was in the 1996 Atlanta Olympic Games. In

But to the dismay of the keen-eyed who wanted the development and evolution of the technology and the actual stakeholders, a series of misidentifications led to the system getting a lot of flak from the public and the media alike.

In the interim, the International Biometric Industry Association (IBIA) was founded in Washington, DC, in the year 1998 and became one of the key bodies in regulating the development and deployment of biometric recognition techniques across the globe. Several parallel developments took place between the 1980s and the early 2000s, bringing the biometric recognition techniques from the science labs to common public platforms and use. And, tracing each of these in the limited space we have would not have been possible, so we had to inevitably skip covering some of those here. From the simple fingerprint on a paper technique to IR blasters packed in a small phone notch that is able to distinguish between two similar faces in milliseconds, we have come a long way.

Looking back at the history of biometrics, unlike many other topics that we have covered in the past and will be covering in the future, you cannot point out one individual or organisation that made biometrics what it is today. It has been a collective effort of a lot of public and private institutions and people working with a lack and a lot of funds to bring the techniques and technology to where it is today. **d**